



**DEPARTMENT OF ARTIFICIAL INTELLIGENCE AND DATA SCIENCE**

|                                                                                                                                                                                                                                                                                                                                                                                                          |                                         |          |          |          |          |
|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------|----------|----------|----------|----------|
| <b>R21UAD502</b>                                                                                                                                                                                                                                                                                                                                                                                         | <b>Formal Languages and Computation</b> | <b>L</b> | <b>T</b> | <b>P</b> | <b>C</b> |
|                                                                                                                                                                                                                                                                                                                                                                                                          |                                         | 3        | 0        | 3        | 3        |
| <b>PRE-REQUISITE:</b>                                                                                                                                                                                                                                                                                                                                                                                    |                                         |          |          |          |          |
| <b>COURSE OBJECTIVES:</b> <ul style="list-style-type: none"><li>● To understand foundations of computation including automata theory</li><li>● To construct models of regular expressions and languages.</li><li>● To design context free grammar and push down automata</li><li>● To design Turing machines and their capability</li><li>● To understand Undecidability and NP class problems</li></ul> |                                         |          |          |          |          |

|                                                                                                                                                                                                                                                                                                                                                                                  |                                                    |   |
|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------|---|
| <b>Unit 1</b>                                                                                                                                                                                                                                                                                                                                                                    | <b>AUTOMATA</b>                                    | 9 |
| Introduction to formal proof – Finite Automata (FA) – Deterministic Finite Automata (DFA) – Non-deterministic Finite Automata (NFA) – Applications of Finite Automata - Equivalence between NFA and DFA – Finite Automata with Epsilon transitions – Minimization of automata.                                                                                                   |                                                    |   |
| <b>Unit 2</b>                                                                                                                                                                                                                                                                                                                                                                    | <b>REGULAR EXPRESSION AND LANGUAGES</b>            | 9 |
| Regular expression – Equivalence of finite automata and regular expressions – Regular Expressions In Lexical Analysis - Proving languages not to be regular (Pumping Lemma) – Closure properties of regular languages.                                                                                                                                                           |                                                    |   |
| <b>Unit 3</b>                                                                                                                                                                                                                                                                                                                                                                    | <b>CONTEXT FREE GRAMMAR AND PUSH DOWN AUTOMATA</b> | 9 |
| Context-Free Grammar (CFG) – Parse trees – Ambiguity in grammars and languages – Definition of the pushdown automata (PDA) – Languages of pushdown automata – Equivalence of pushdown automata and CFG – Deterministic Pushdown Automata                                                                                                                                         |                                                    |   |
| <b>Unit 4</b>                                                                                                                                                                                                                                                                                                                                                                    | <b>NORMAL FORMS AND TURING MACHINES</b>            | 9 |
| Normal forms for CFG – Chomsky Normal Form and Greibach Normal Form – Pumping lemma for CFL – Closure properties of Context Free Languages. Turing machines: Basic model – definition and representation – Instantaneous Description – Language acceptance by TM – Variants of Turing Machine – TM as Computer of Integer functions – Programming techniques for Turing machines |                                                    |   |
| <b>Unit 5</b>                                                                                                                                                                                                                                                                                                                                                                    | <b>UNDECIDABILITY</b>                              | 9 |
| Recursively Enumerable Language (RE) – An undecidable problem that is RE – Undecidable problems about Turing Machine – Post’s correspondence problem – The classes P and NP – Kruskal’s algorithm – Traveling Salesman Problem.                                                                                                                                                  |                                                    |   |
| <b>Total periods: 45 periods</b>                                                                                                                                                                                                                                                                                                                                                 |                                                    |   |

**Course Outcomes**

*At the end of the course the student will be able to*



|            |  |  |          |  |          |  |  |  |          |          |          |          |          |
|------------|--|--|----------|--|----------|--|--|--|----------|----------|----------|----------|----------|
| <b>CO5</b> |  |  | <b>3</b> |  | <b>3</b> |  |  |  |          |          |          | <b>3</b> |          |
| <b>CO6</b> |  |  |          |  |          |  |  |  | <b>2</b> | <b>2</b> | <b>2</b> | <b>2</b> | <b>2</b> |



# SETHU INSTITUTE OF TECHNOLOGY, KARIAPATTI

(An Autonomous Institution Affiliated to Anna University, Chennai)

Regulation R2021 (Question Bank)

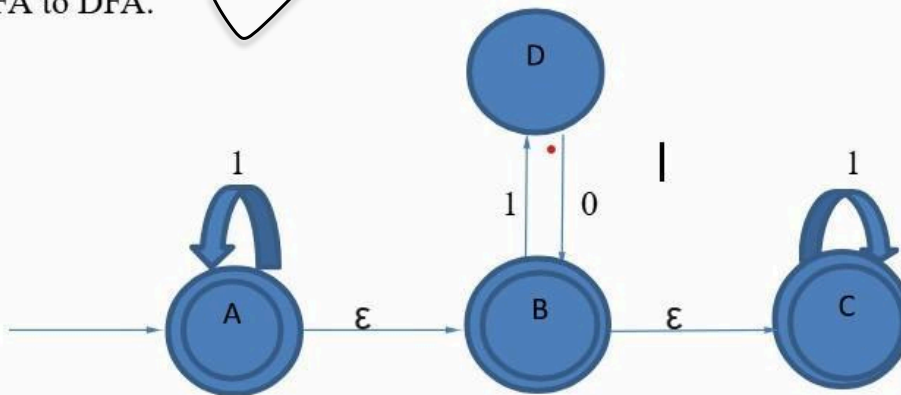
|                                                              |                                                                                                                                             |                                                        |
|--------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------|
| <b>Department: Artificial Intelligence and Data Science</b>  |                                                                                                                                             | <b>Subject Name : Formal Languages and Computation</b> |
| <b>Subject code : R21UAD502</b>                              |                                                                                                                                             | <b>Question Pattern :10*2=20<br/>5*16=80</b>           |
| <b>Course Coordinator : Mr. S.Narendran<br/>AP\AI&amp;DS</b> |                                                                                                                                             | <b>Time Duration : 3 Hours</b>                         |
| PART – A ( 2 Marks)                                          |                                                                                                                                             |                                                        |
| UNIT - I (Minimum 8 Questions)                               |                                                                                                                                             |                                                        |
| 1.                                                           | Define Deterministic Finite Automaton.                                                                                                      | CO1- U                                                 |
| 2.                                                           | State any four types of proofs.                                                                                                             | CO1- U                                                 |
| 3.                                                           | Create a FA which checks whether the given binary number is even.                                                                           | CO1- U                                                 |
| 4.                                                           | Summarize minimization of DFA                                                                                                               | CO1- U                                                 |
| 5.                                                           | Construct the non-deterministic automata to accept strings containing the substring 0101.                                                   | CO2- AP                                                |
| 6.                                                           | Create a FA which accepts the only input 101 over the input set $Z=\{0,1\}$                                                                 | CO2- AP                                                |
| 7.                                                           | Summarize the extended transition function for a $\epsilon$ -NFA                                                                            | CO1- U                                                 |
| 8.                                                           | Find DFA over $\Sigma=(a,b)$ which produces not more than $3a$ 's.                                                                          | CO2- AP                                                |
| UNIT - II (Minimum 8 Questions)                              |                                                                                                                                             |                                                        |
| 1.                                                           | State pumping lemma and its advantages.                                                                                                     | CO1- U                                                 |
| 2.                                                           | Find whether a language $L=(0^n 1^{2n}   n>0)$ is regular or not using pumping lemma                                                        | CO2- AP                                                |
| 3.                                                           | Compare the difference between the + closure and * closure                                                                                  | CO1- U                                                 |
| 4.                                                           | Find the finite automaton for the regular expression $0^*1^*$ .                                                                             | CO2-AP                                                 |
| 5.                                                           | Find a DFA for the regular expression $aa^*bb^*$ .                                                                                          | CO2- AP                                                |
| 6.                                                           | Construct a regular expression to recognize the set of strings over $\Sigma= \{a,b\}$ having odd number of a's and b's and starts with 'a'. | CO2- AP                                                |
| 7.                                                           | Name the four closure properties of RE.                                                                                                     | CO1- U                                                 |
| 8.                                                           | Find out the language generated by the regular expression $(0+1)^*$ .                                                                       | CO2- AP                                                |
| UNIT - III (Minimum 8 Questions)                             |                                                                                                                                             |                                                        |
| 1.                                                           | Construct context free grammar $\{0^m 1^n   1 \leq m \leq n\}$                                                                              | CO2- AP                                                |
| 2.                                                           | What are the ways of language acceptance in PDA.                                                                                            | CO1- U                                                 |
| 3.                                                           | Define Push Down Automata                                                                                                                   | CO1- U                                                 |
| 4.                                                           | Explain ambiguous with example?                                                                                                             | CO1- U                                                 |
| 5.                                                           | Find PDA to accept the strings of the language, $L=\{a^n c^m b^n, n,m \geq 0\}$                                                             | CO2- AP                                                |
| 6.                                                           | Define the pumping Lemma for CFL.                                                                                                           | CO1- U                                                 |

|                                                 |                                                                                                                                                                                                                                |             |
|-------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------|
| 7.                                              | Find the string aaabbabbba for the Grammar G with                                                                                                                                                                              |             |
|                                                 | $S \rightarrow aB bA$ $A \rightarrow a aS bAA$ $B \rightarrow b bS aBB$                                                                                                                                                        | CO2- AP     |
| 8.                                              | Find derivation tree for the string abaaba for the CFG given by, G where Productions are $S \rightarrow aSa$ $S \rightarrow bSb$ $S \rightarrow a b  \epsilon$                                                                 | CO2- AP     |
| UNIT - IV (Minimum 8 Questions)                 |                                                                                                                                                                                                                                |             |
| 1.                                              | What are the two normal forms of CFG? Write their productions format.                                                                                                                                                          | CO1-U       |
| 2.                                              | Define the language recognized by any Turing Machine                                                                                                                                                                           | CO1-U       |
| 3.                                              | Find a transition diagram for a Turing machine to identify $n \bmod 2$ .                                                                                                                                                       | CO2- AP     |
| 4.                                              | Show the following grammar into an equivalent one with no unit productions and no useless symbols<br>$S \rightarrow ABA$ $A \rightarrow aAaCaBC bB$ $B \rightarrow A bB Cb$ $C \rightarrow CC Cc$                              | CO2- AP     |
| 5.                                              | Find $L(G)$ , where $G = \{\{S\}, \{0, 1\}, \{S \rightarrow 0S1, S \rightarrow \epsilon\}, S\}$                                                                                                                                | CO2- AP     |
| 6.                                              | Find a grammar $G'$ in CNF form equivalent G<br>$S \rightarrow aAD$ , $A \rightarrow aB baB$ , $B \rightarrow b$ , $D \rightarrow d$ .                                                                                         | CO2- AP     |
| 7.                                              | Eliminate the $\epsilon$ productions from the CFG given below<br>$A \rightarrow 0B1 1B1$ $B \rightarrow 0B 1B  \epsilon$                                                                                                       | CO2- AP     |
| 8.                                              | Find CFG with no useless symbols equivalent to<br>$S \rightarrow AB CA$ $A \rightarrow aB BC AB$ $C \rightarrow aB b$                                                                                                          | CO2- AP     |
| UNIT - V (Minimum 8 Questions)                  |                                                                                                                                                                                                                                |             |
| 1.                                              | Define recursive languages?                                                                                                                                                                                                    | CO1- U      |
| 2.                                              | Define multitape turing machine                                                                                                                                                                                                | CO1- U      |
| 3.                                              | Find TM for $L = \{a^n b^n c^n \mid n \geq 0\}$                                                                                                                                                                                | CO2- AP     |
| 4.                                              | Find Turing machine with not more than three states that accepts the language $a(a+b)^*$ . Assume $\Sigma = \{a, b\}$                                                                                                          | CO2- AP     |
| 5.                                              | Describe an example of an undecidable problem                                                                                                                                                                                  | CO1- U      |
| 6.                                              | Define a universal language $L_u$ ?                                                                                                                                                                                            | CO1- U      |
| 7.                                              | What is meant by halting problem                                                                                                                                                                                               | CO1 – U     |
| 8.                                              | What are the difference between finite automata and turing machine                                                                                                                                                             | CO1 –U      |
| PART – B                                        |                                                                                                                                                                                                                                |             |
| UNIT - I (Minimum 10 Questions without choice ) |                                                                                                                                                                                                                                |             |
| 1.                                              | Construct the NFA that accepts all strings that end in 01. Give its transition table and the Extended transition function for the input string 00101. Also construct a DFA for the above NFA using subset construction method. | CO2-AP (16) |

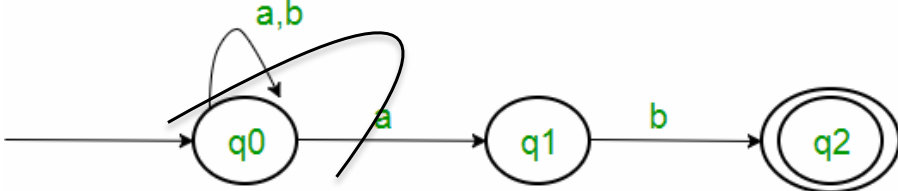
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|----|-------------------------------------------------------------------------------------------------------------------------------------|--------|------|
| 2. | Construct DFA and find DFA equivalent to the given NFA ( $\{p, q\}, \{0, 1\}, \delta, \{p\}, \{s\}$ ), where $\delta$ is defined as | CO2-AP | (16) |
| 3. |                                                                                                                                     | CO2-AP | (16) |

|    | 0     | 1      |
|----|-------|--------|
| p  | {p,q} | {p}    |
| q  | {r}   | {r}    |
| r  | {s}   | $\phi$ |
| *s | {s}   | {s}    |

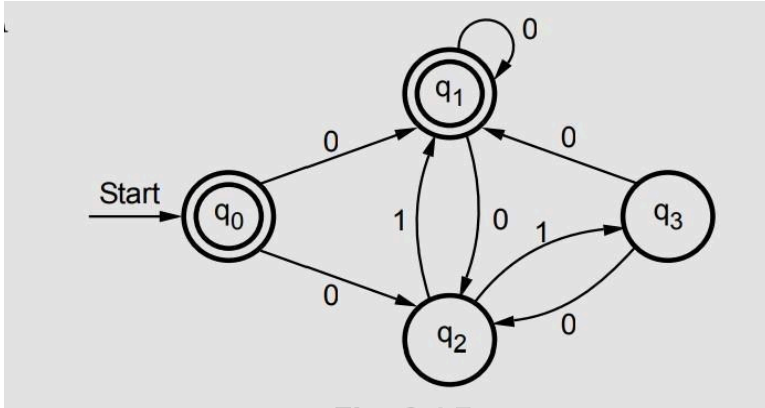
Convert the following  $\epsilon$ -NFA to NFA and then convert the resultant NFA to DFA.



|    |                                                                                                                          |        |      |
|----|--------------------------------------------------------------------------------------------------------------------------|--------|------|
| 4. | <b>Identify</b> and explain the algorithm for minimization of DFA. Using the above algorithm minimize the following DFA. | CO2-AP | (16) |
|    |                                                                                                                          |        |      |

|    |                                                                                                                                                         |        |      |
|----|---------------------------------------------------------------------------------------------------------------------------------------------------------|--------|------|
| 5. | <p>(i) Construct the following NFA to DFA.</p> <p>(ii) Construct a DFA accepting binary strings such that the third symbol from the right end is 1.</p> | CO2-AP | (16) |
| 6. | <p>Construct DFA equivalent to the NFA given below.</p>               | CO2-AP | (16) |
| 7. | <p>(i) Convert the following NFA into DFA</p>                                                                                                           | CO2-AP | (16) |

| States | a      | b      |
|--------|--------|--------|
| P      | {p}    | {p, q} |
| q      | {r}    | {r}    |
| r*     | $\phi$ | $\phi$ |

|    |                                                                                                                                                                                                                                      |        |      |
|----|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------|------|
|    |  <p>(ii) Construct a NFA to accepting the set of strings over {a,b} ending in aba. Use it construct a DFA accepting the same set of strings.</p> |        |      |
| 8. | <p>Construct the following <math>\epsilon</math>-NFA to DFA.</p>                                                                                                                                                                     | CO2-AP | (16) |

|     |                                                                                                                                                                                                                                                                                         |        |      |
|-----|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------|------|
| 9.  | Construct an NFA that accepts all strings that end with the substring "01".<br>a. Construct the DFA states and transitions based on the subsets of NFA states.<br>b. Provide the transition table for the resulting DFA<br>c. Identify the start state and the accept states of the DFA | CO1-U  | (16) |
| 10. | Explain and difference between the NFA and DFA. Convert the following $\epsilon$ -NFA to DFA.                                                                                                                                                                                           | CO1-U  | (16) |
| 11. | (i) Draw and explain the transition diagram for recognizing the set of all operators in C language.<br>(ii) Find a DFA from the given NFA. $M = (\{q_0, q_1\}, \{a, b\}, \delta, q_0, \{q_1\})$ with the state table diagram for $\delta$ given below:                                  | CO2-AP | (16) |
| 12. | (i) Define Deterministic Finite Automaton, Non-Deterministic Finite Automaton, its components and how it operates.<br>(ii) Explain how a DFA differs from a Non-deterministic Finite Automaton                                                                                          | CO1-U  | (16) |

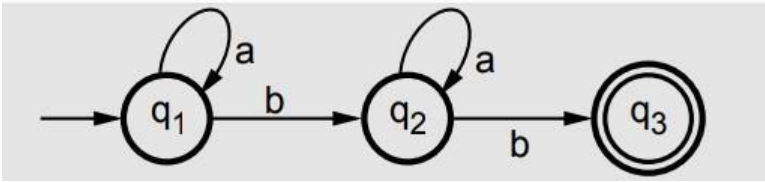
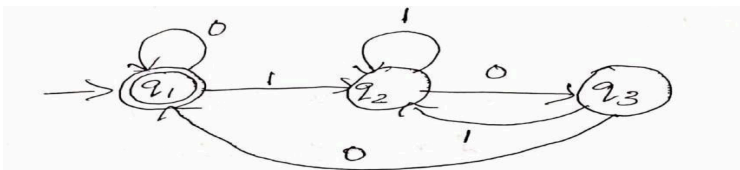
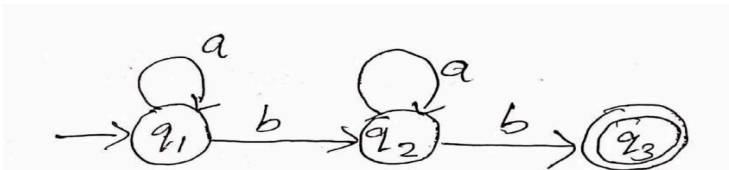
| states | $\epsilon$ | a   | b      | c      |
|--------|------------|-----|--------|--------|
| p      | $\Phi$     | {p} | {q}    | {r}    |
| q      | {p}        | {q} | {r}    | {p, q} |
| *r     | {q}        | {r} | $\phi$ | {p}    |

| states | $\epsilon$ | a   | b      | c      |
|--------|------------|-----|--------|--------|
| p      | $\Phi$     | {p} | {q}    | {r}    |
| q      | {p}        | {q} | {r}    | $\Phi$ |
| *r     | {q}        | {r} | $\phi$ | {p}    |

| states | a      | b      |
|--------|--------|--------|
| P      | {p, q} | q      |
| q      | $\phi$ | {p, q} |

|                                                                                     |                                                                     |       |      |
|-------------------------------------------------------------------------------------|---------------------------------------------------------------------|-------|------|
|  | UNIT - II (Minimum 10 Questions <b>without choice</b> )             |       |      |
| 1.                                                                                  | Explain and prove any four closure properties of Regular Languages. | CO1-U | (16) |



|    |                                                                                                                                                                                                                                                                                          |        |      |
|----|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------|------|
| 2. | Construct NFA for given RE using Thomson rule.<br>i) $a.(a+b)^* ab$<br>ii) $(a.b)^*$<br>iii) $(a+b)$                                                                                                                                                                                     | CO2-AP | (16) |
| 3. | Prove that the following languages are not regular<br>(i) $\{w \in \{a, b\}^* \mid w = WR\}$<br>(ii) Find NFA equivalent to the regular expression $(0+1)^*(00+11)(0+1)^*$                                                                                                               | CO2-AP | (16) |
| 4. | (i) Find the regular expression for finite automata<br><br>(ii) Construct a finite automata for the regular expression $10+(0+11)0^*1$                                                                 | CO2-AP | (16) |
| 5. | (i) Find the given language $L = \{0^{2n} \mid n \geq 1\}$ is regular or not.<br>(ii) Find the given language $L = \{b^{i^2} \mid i \geq 1\}$ is regular or not.                                                                                                                         | CO2-AP | (16) |
| 6. | (i) Construct an NFA equivalent to the regular expression $(0+1)^*(00+11)(0+1)^*$ .<br>(ii) Construct a finite automata for the regular expression $10+(0+11)0^*1$                                                                                                                       | CO2-AP | (16) |
| 7. | Construct the regular expression corresponding to the statediagram given in the following<br><br>(i)<br><br>(ii) | CO2-AP | (16) |
| 8. | (i) Using pumping lemma for regular sets, prove that the language $L =$                                                                                                                                                                                                                  | CO2-AP | (16) |

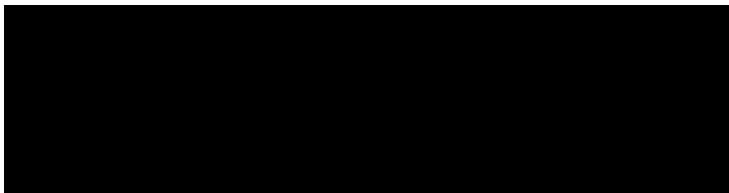
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|--------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------|------|
|                                                  | $\{0^m 1^n 0^{m+n} \mid m \geq 1 \text{ and } n \geq 1\}$ is not regular.<br>(ii) find the given language $L = \{a^n b^m \mid n, m \geq 1\}$ is regular or not.                                                                                                                                                                                                                                                                                                                                                         |        |      |
| 9.                                               | Prove that regular languages are closed under each of these operations. For your proofs, use the following approach: <ul style="list-style-type: none"> <li><b>Union:</b> Given two regular languages <math>L_1</math> and <math>L_2</math>, show that their union <math>L_1 \cup L_2</math> is also a regular language.</li> <li><b>Intersection:</b> Given two regular languages <math>L_1</math> and <math>L_2</math>, show that their intersection <math>L_1 \cap L_2</math> is also a regular language.</li> </ul> | CO1-U  | (16) |
| 10.                                              | i) Prove the $L_1$ and $L_2$ are two languages then $L_1 \cup L_2$ is regular<br>ii) Prove the $L_1$ and $L_2$ are two languages then $L_1 \cap L_2$ is regular                                                                                                                                                                                                                                                                                                                                                         | CO1-U  | (16) |
| 11.                                              | Define and Explain the closure properties of regular languages with respect to the following operations: <ul style="list-style-type: none"> <li>Union</li> <li>Intersection</li> <li>Complement</li> <li>Concatenation</li> <li>Kleene Star</li> </ul>                                                                                                                                                                                                                                                                  | CO1-U  | (16) |
| 12.                                              | (i) Find the equalities for the following RE and prove for the same <ol style="list-style-type: none"> <li><math>b+ab^*+aa^*b+aa^*ab^*</math></li> <li><math>a^*(b+ab^*)</math>.</li> <li><math>a(a+b)^*+aa(a+b)^*+aaa(a+b)^*</math></li> </ol> (ii) A homomorphism of regular language is regular                                                                                                                                                                                                                      | CO2-AP | (16) |
| UNIT - III (Minimum 10 Questions without choice) |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |        |      |
| 1.                                               | Construct the grammar for the following PDA $PDA = (\{q_0, q_1\}, \{0, 1\}, \{X, Z_0\}, \delta, q_0, Z_0, \Phi)$ and where $\delta$ is given by $\begin{aligned} \delta(q_0, 1, Z_0) &= \{(q_0, XZ_0)\}, \\ \delta(q_0, 1, X) &= \{(q_0, XX)\}, \\ \delta(q_0, 0, X) &= \{(q_1, X)\}, \\ \delta(q_0, \epsilon, Z_0) &= \{(q_0, \epsilon)\}, \\ \delta(q_1, 1, X) &= \{(q_1, \epsilon)\}, \\ \delta(q_1, 0, Z_0) &= \{(q_0, Z_0)\}. \end{aligned}$                                                                       | CO1-U  | (16) |
| 2.                                               | (i) Construct PDA for the given CFG $S \rightarrow aSb \quad S \rightarrow ab$<br>(ii) Construct PDA for the following grammar $S \rightarrow AB \quad A \rightarrow CD \quad B \rightarrow b \quad C \rightarrow a \quad D \rightarrow a$                                                                                                                                                                                                                                                                              | CO1-U  | (16) |
| 3.                                               | Solve the following grammar<br>$S \rightarrow aAa \mid bBb \mid BB$<br>$A \rightarrow C \mid B \mid S$<br>$A \rightarrow \epsilon$<br>$C \rightarrow S \mid \epsilon$ for the string abaaba. Find<br>i) Left most derivation<br>ii) Right most derivation<br>iii) Derivation Tree<br>iv) For the string abaabbba, find the right most derivation.                                                                                                                                                                       | CO2-AP | (16) |

|     |                                                                                                                                                                                                                                                                                                                                                                                                                                                      |        |      |
|-----|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------|------|
| 4.  | Find PDA that recognizes and analyzes the language $\{a^i b^j c^k \mid i, j, k > 0 \text{ and } i=j \text{ or } i=k\}$ . Explain about PDA acceptance                                                                                                                                                                                                                                                                                                | CO1-U  | (16) |
|     | <ul style="list-style-type: none"> <li>From empty Stack to final state.</li> <li>From Final state to Empty Stack.</li> </ul>                                                                                                                                                                                                                                                                                                                         |        |      |
| 5.  | Find CFG G which accepts $N(M)$ , where $M = (\{q_0, q_1\}, \{a, b\}, \{z_0, z\}, \delta, q_0, z_0, \Phi)$ and where $\delta$ is given by $\begin{aligned} \delta(q_0, b, z_0) &= \{(q_0, zz_0)\} \\ \delta(q_0, \epsilon, z_0) &= \{(q_0, \epsilon)\} \\ \delta(q_0, b, z) &= \{(q_0, zz)\} \\ \delta(q_0, a, z) &= \{(q_1, z)\} \\ \delta(q_1, b, z) &= \{(q_1, \epsilon)\} \\ \delta(q_1, a, z_0) &= \{(q_0, z_0)\} \end{aligned}$                | CO1-U  | (16) |
| 6.  | Construct a PDA accepting $\{a^n b^m a^n \mid m, n \geq 1\}$ by empty stack. Also find the corresponding CFG accepting for the same set.                                                                                                                                                                                                                                                                                                             | CO1-U  | (16) |
| 7.  | Consider the following grammar for productions:<br>$S \rightarrow a \mid \wedge \mid (T) \quad T \rightarrow T, S \mid S$ . Find the leftmost derivation and rightmost derivation and parse tree for $((a, a), (\wedge, a), a)$                                                                                                                                                                                                                      | CO2-AP | (16) |
| 8.  | (i) Find PDA for the given CFG and analyze the answer $(a^+ b)$ and $a^{++}$ .<br>$I \rightarrow a \mid b \mid a \mid b \mid I \mid I$<br>$E \rightarrow I \mid E + E \mid E * E \mid (E)$<br>(iii) Find PDA for the grammar $S \rightarrow aSb \mid A, A \rightarrow bSa \mid S \mid \epsilon$ that accepts the language by empty stack.                                                                                                            | CO1-U  | (16) |
| 9.  | Consider the grammar,<br>$S \rightarrow iCtS$<br>$S \rightarrow iCtSeS$<br>$S \rightarrow a$<br>$C \rightarrow b$ . Where i, t, e, C & S stand for if, else, then, “Conditional” and “Statement” respectively.<br>(i) Construct leftmost derivation for the sentence $W = ibtibtaea$ .<br>(ii) Show the corresponding parse tree for the above sentence Find the above grammar is ambiguous. If yes remove ambiguity.                                | CO1-U  | (16) |
| 10. | (i) Explain acceptance by final state and acceptance by empty stack of the push down automata.<br>(ii) Convert PDA to CFG. PDA is given by $P = (\{p, q\}, \{0, 1\}, \{X, Y\}, \delta, q, Z)$ is defined by $\delta(p, 1, z) = \{(p, XZ)\}$ ,<br>$\delta(p, \epsilon, Z) = \{(p, \epsilon)\}$ ,<br>$\delta(p, 1, X) = \{(p, XX)\}$ ,<br>$\delta(q, 1, X) = \{(q, \epsilon)\}$ ,<br>$\delta(p, 0, X) = \{(q, X0)\}$<br>$\delta(q, 0, Z) = \{(p, Z)\}$ | CO2-AP | (16) |
| 11. | Define a Context-Free Grammar (CFG) and a Pushdown Automaton (PDA). Differentiate the relationship between CFGs and PDAs                                                                                                                                                                                                                                                                                                                             | CO1-U  | (16) |

|     |                                                                                                                                                                                                                                                                                                                                        |       |      |
|-----|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------|------|
| 12. | Design a Pushdown Automaton for the language L defined as: $L = \{a^n b^m c^k \mid n=m \text{ or } m=k \text{ or } n=k\}$ . Explain how the PDA handles the three different conditions of the language LL. For each condition (i.e., $n=m$ , $m=k$ , and $n=k$ ), describe how the PDA processes the input and makes use of its stack. | CO1-U | (16) |
|     | UNIT - IV (Minimum 10 Questions <b>without</b> choice )                                                                                                                                                                                                                                                                                |       |      |

|                                                 |                                                                                                                                                                                                                                                                                                                                                                                            |         |        |        |   |       |       |   |   |     |   |       |    |   |    |   |        |      |
|-------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------|--------|--------|---|-------|-------|---|---|-----|---|-------|----|---|----|---|--------|------|
| 1.                                              | Explain the conversion from CFG into CNF accepting the language?<br>Convert the following CFG into GNF.<br>$S \rightarrow b A   a B$<br>$A \rightarrow b A A   a S   a B$<br>$\rightarrow a B B   b S   b$                                                                                                                                                                                 | CO1- AP | (16)   |        |   |       |       |   |   |     |   |       |    |   |    |   |        |      |
| 2.                                              | (i) Explain two types of normal form and gives an example.<br>(ii) Explain the null productions, Unit production and Useless symbols in production with example.                                                                                                                                                                                                                           | CO1- U  | (16)   |        |   |       |       |   |   |     |   |       |    |   |    |   |        |      |
| 3.                                              | (i) Define pumping lemma for CFL. Show that $L = \{a^i b^j c^k, i < j < k\}$ is not context free.<br>(ii) Find a grammar in Chomsky normal form equivalent to $S \rightarrow aAbB, A \rightarrow aA a, B \rightarrow bB b$ .                                                                                                                                                               | CO2-AP  | (16)   |        |   |       |       |   |   |     |   |       |    |   |    |   |        |      |
| 4.                                              | (i) Find a grammar in Chomsky normal form equivalent to $S \rightarrow aAD, A \rightarrow aB bAB, B \rightarrow b, D \rightarrow d$ .<br>(ii) Find a grammar in Greibach normal form equivalent to $S \rightarrow AA 0, A \rightarrow SS 1$                                                                                                                                                | CO2-AP  | (16)   |        |   |       |       |   |   |     |   |       |    |   |    |   |        |      |
| 5.                                              | Find grammar G into Greibach Normal Form (GNF)<br>$S \rightarrow XA BB, B \rightarrow b SB, X \rightarrow b, A \rightarrow a$                                                                                                                                                                                                                                                              | CO2-AP  | (16)   |        |   |       |       |   |   |     |   |       |    |   |    |   |        |      |
| 6.                                              | (i) Find GNF for the given grammar $S \rightarrow abSb, S \rightarrow aa$ .<br>(ii) Find CNF to the following grammar $S \rightarrow cBA, S \rightarrow A, A \rightarrow cB AbbS, B \rightarrow aaa$ .                                                                                                                                                                                     | CO2-AP  | (16)   |        |   |       |       |   |   |     |   |       |    |   |    |   |        |      |
| 7.                                              | (i) Find TM for language consisting of strings having any number of 0's and only even numbers of 1's over the input set $= \{0, 1\}$<br>(ii) Find TM for $L = \{a^n b^n c^n   n \geq 0\}$                                                                                                                                                                                                  | CO2-AP  | (16)   |        |   |       |       |   |   |     |   |       |    |   |    |   |        |      |
| 8.                                              | Find the Turing machine for multiplication using the subroutine 'copy'.                                                                                                                                                                                                                                                                                                                    | CO2-AP  | (16)   |        |   |       |       |   |   |     |   |       |    |   |    |   |        |      |
| 9.                                              | Find TM for reversing a binary string on the input tape.                                                                                                                                                                                                                                                                                                                                   | CO2-AP  | (16)   |        |   |       |       |   |   |     |   |       |    |   |    |   |        |      |
| 10.                                             | (i) Find TM to compute $n^2$ .<br>(ii) Find multitape TM for checking whether given number is prime.                                                                                                                                                                                                                                                                                       | CO2-AP  | (16)   |        |   |       |       |   |   |     |   |       |    |   |    |   |        |      |
| 11.                                             | Explain Chomsky Normal Form and Greibach Normal Form and explain the Null Productions, Unit productions and Useless symbols.                                                                                                                                                                                                                                                               | CO1- U  | (16)   |        |   |       |       |   |   |     |   |       |    |   |    |   |        |      |
| 12.                                             | Define and prove the any four closure properties of context free languages                                                                                                                                                                                                                                                                                                                 | CO1- U  | (16)   |        |   |       |       |   |   |     |   |       |    |   |    |   |        |      |
| UNIT - V (Minimum 10 Questions without choice ) |                                                                                                                                                                                                                                                                                                                                                                                            |         |        |        |   |       |       |   |   |     |   |       |    |   |    |   |        |      |
| 1.                                              | Explain post correspondence problem .Let $\Sigma = \{0,1\}$ . Let A and B be the lists of three strings each ,defined<br><table><tr><td></td><td>List A</td><td>List B</td></tr><tr><td>i</td><td><math>w_i</math></td><td><math>x_i</math></td></tr><tr><td>1</td><td>1</td><td>111</td></tr><tr><td>2</td><td>10111</td><td>10</td></tr><tr><td>3</td><td>10</td><td>0</td></tr></table> |         | List A | List B | i | $w_i$ | $x_i$ | 1 | 1 | 111 | 2 | 10111 | 10 | 3 | 10 | 0 | CO2-AP | (16) |
|                                                 | List A                                                                                                                                                                                                                                                                                                                                                                                     | List B  |        |        |   |       |       |   |   |     |   |       |    |   |    |   |        |      |
| i                                               | $w_i$                                                                                                                                                                                                                                                                                                                                                                                      | $x_i$   |        |        |   |       |       |   |   |     |   |       |    |   |    |   |        |      |
| 1                                               | 1                                                                                                                                                                                                                                                                                                                                                                                          | 111     |        |        |   |       |       |   |   |     |   |       |    |   |    |   |        |      |
| 2                                               | 10111                                                                                                                                                                                                                                                                                                                                                                                      | 10      |        |        |   |       |       |   |   |     |   |       |    |   |    |   |        |      |
| 3                                               | 10                                                                                                                                                                                                                                                                                                                                                                                         | 0       |        |        |   |       |       |   |   |     |   |       |    |   |    |   |        |      |

|    |                                                                                            |       |      |
|----|--------------------------------------------------------------------------------------------|-------|------|
|    | (i) Does the PCP have a solution?<br>(ii) Prove that the universal language is recursively |       |      |
| 2. | Describe about the tractable and intractable problems with an examples.                    | CO1-U | (16) |
| 3. | Explain PCP and prove that PCP is undecidable.                                             | CO1-U | (16) |



|               |                                                                                                                                                                                             |        |      |
|---------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------|------|
| 4.            | Describe about Recursive and Recursive Enumerable languages with examples.                                                                                                                  | CO1-U  | (16) |
| 5.            | (i) Describe about the tractable and intractable problems.<br>(ii) Explain that "MPCP reduce to PCP".                                                                                       | CO1-U  | (16) |
| <del>6.</del> | Find how a multi-track in a TM can be used for testing given positive integer is a prime or not.                                                                                            | CO2-AP | (16) |
| 7             | (i) Construct a Turing machine that estimate unary multiplication (Say $111 \times 11 = 11111$ )<br>(ii) Does PCP with two lists $x=(b, b ab^3, ba)$ and $y=(b^3, ba, a)$ have a solution.  | CO2-AP | (16) |
| 8.            | (i) Plan and explain on decidable and un-decidable problems with an example.<br>(ii) Design and prove that for two recursive languages L1 and L2 their union and intersection is recursive. | CO1-U  | (16) |
| 9.            | (i) Explain about tractable and intractable problems with an example.<br>(ii) Explain the limitation of automata for Type 3, Type 2, type0 languages.                                       | CO1-U  | (16) |
| 10.           | (i) Define PCP and prove that PCP is undecidable.<br>(ii) Explain that "MPCP reduce to PCP".                                                                                                | CO1-U  | (16) |
| 11.           | Explain and state RICE theorem                                                                                                                                                              | CO1-U  | (16) |
| <del>12</del> | Define a TM to accept the language $LE=\{1^n 2^n 3^n   n \geq 1\}$                                                                                                                          | CO2-AP | (16) |